

MiT4SL: Multi-omics Triplet Representation Learning for Cancer Cell Line-adapted Prediction of Synthetic Lethality

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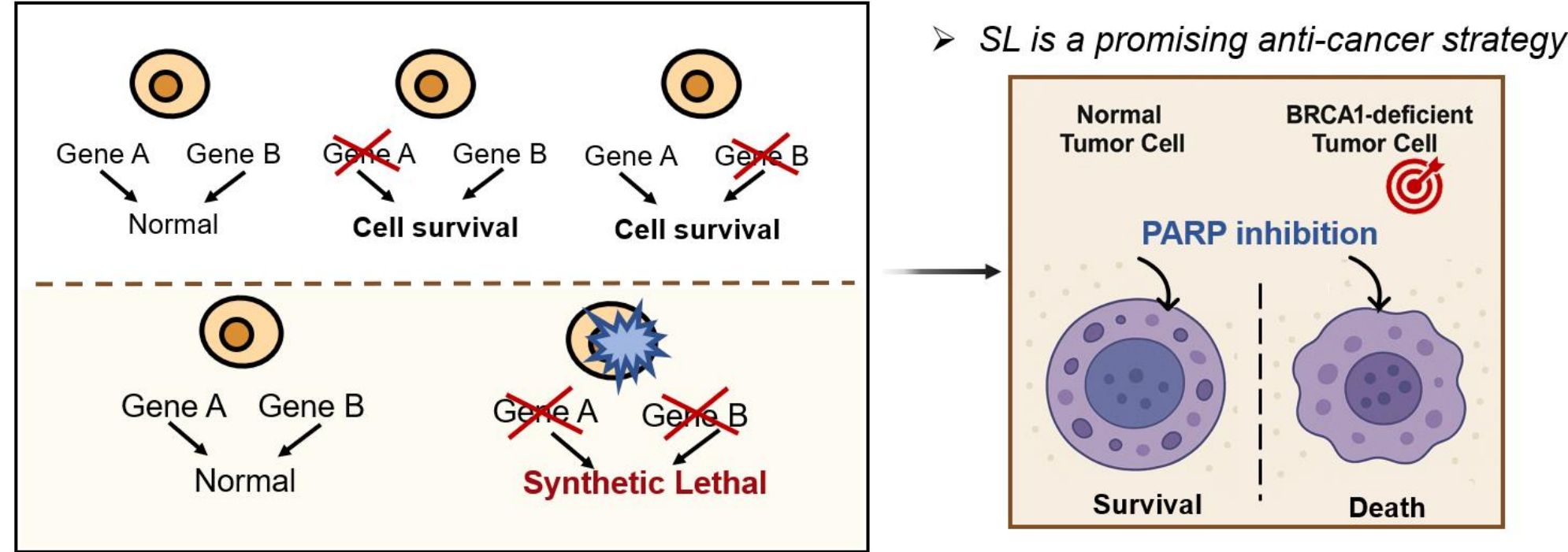
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Introduction

Background

➤ Synthetic lethality (SL):

- Perturbation of one gene → Well-tolerated
- Perturbation of both genes → Cell death



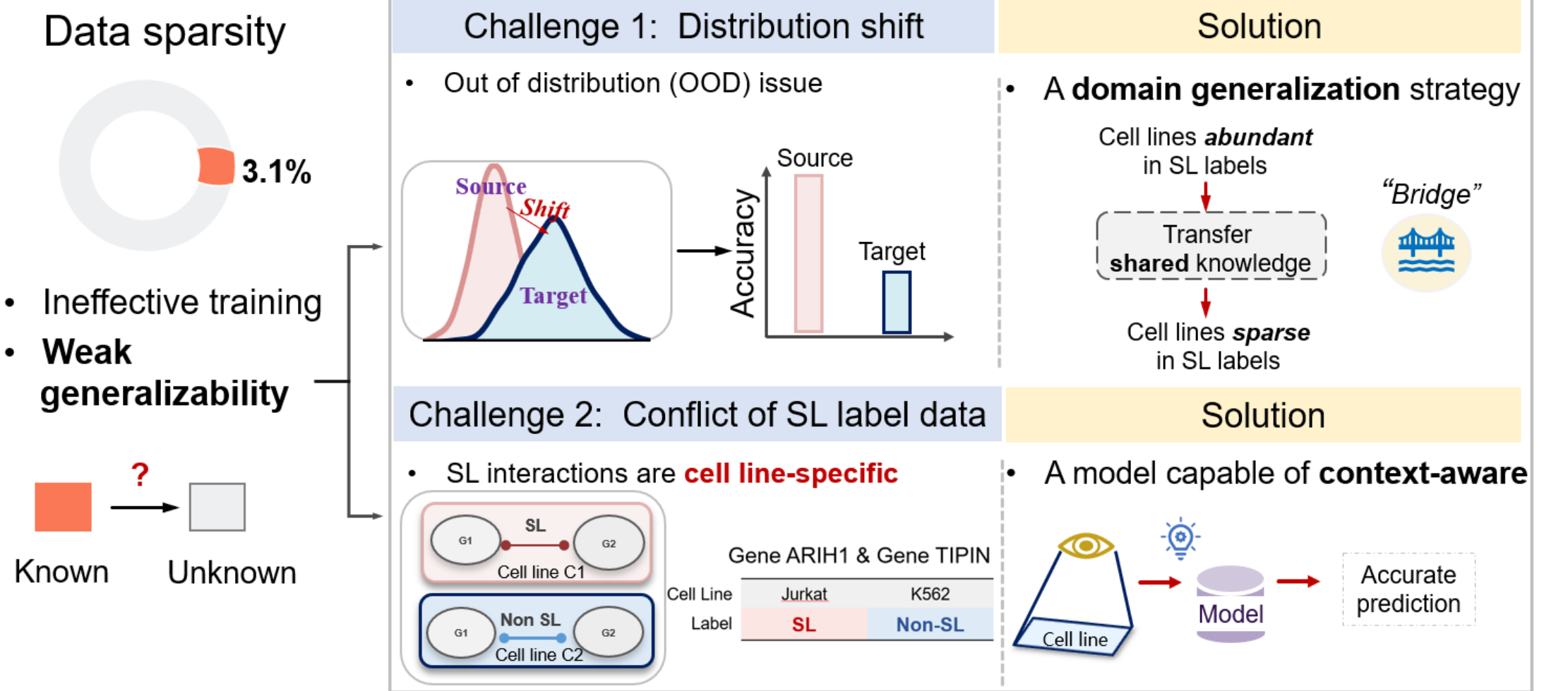
Challenge

- Data sparsity: Only 31 cell lines (~3.1%) have SL label
- Ineffective training
- Weak generalizability

Solution

- A domain generalization-based strategy
- A model capable of context-aware

A robust and context-aware model is essential for accurate synthetic lethality prediction.

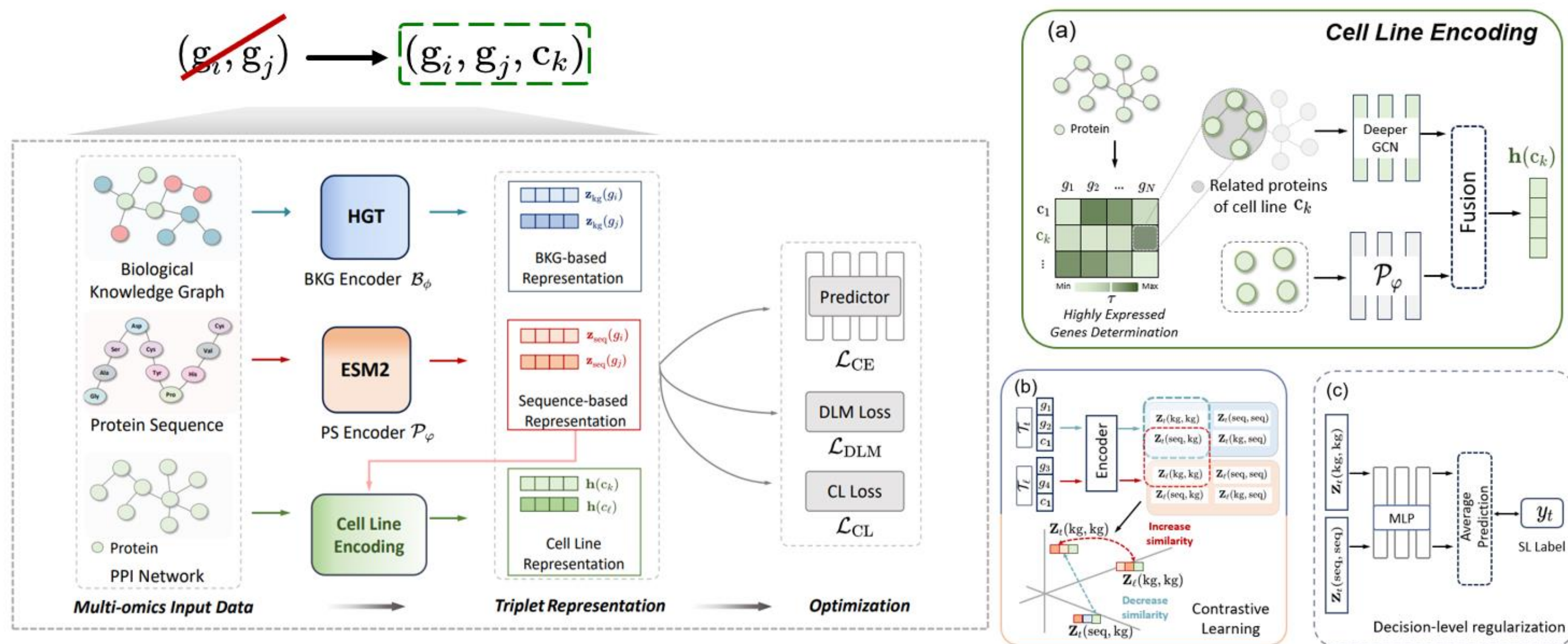


Methodology

MiT4SL: A Cell Line-Adapted Model

- Triplet representation framework
- A cell line encoding strategy
- Dual regularization

- ✓ Shared biological mechanisms
- ✓ Cell line-specific information

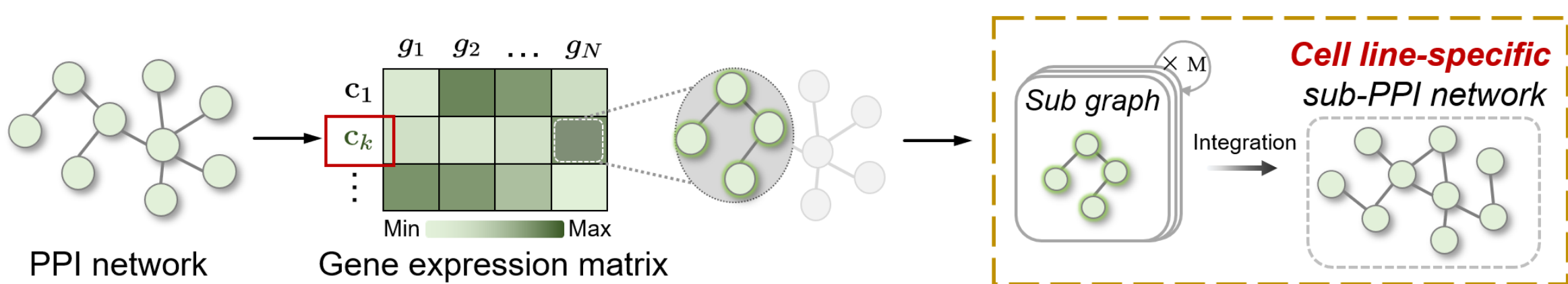


Introducing cell line representation ----> Triplet representation ----> SL prediction

(a) Cell Line Encoding Strategy

- Construction of cell line-specific protein-protein interaction (PPI) network.

Given a cell line, we first determine its corresponding gene set based on a gene expression threshold. Then, we sample the local subgraph for each gene in this set, from the full PPI network. By integrating all these subgraphs, a cell line-specific sub-PPI network is constructed.



- Multi-modality cell line representations

We generate the cell line representations based on two modalities: graph-level and sequence level. The process is defined as:

$$\begin{aligned} \mathbf{z}_{\text{ppi}}(\tilde{g}_c) &= \text{DeeperGCN}(\tilde{g}_c) \\ \mathbf{z}_{\text{seq}}(\tilde{g}_c) &= \mathcal{P}_\varphi(\tilde{g}_c) \\ \mathbf{h}(c) &= \text{Concat}[\mathbf{h}_{\text{ppi}}(c), \mathbf{h}_{\text{seq}}(c)] \end{aligned}$$

(b) Gene Representation

- Gene representation is obtained by combining two complementary data: a biomedical knowledge graph (BKG) and protein sequence. The process can be expressed as :

$$\begin{aligned} \mathbf{z}_{\text{kg}}(g) &= \mathcal{B}_\phi(g) \\ \mathbf{z}_{\text{seq}}(g) &= \mathcal{P}_\varphi(s_g) \end{aligned}$$

(c) Optimization

- MiT4SL is optimized by three loss components: SL prediction score $\mathcal{L}_{\text{CE}}$, contrastive learning-based $\mathcal{L}_{\text{CL}}$, and decision-level loss $\mathcal{L}_{\text{DL}}$.

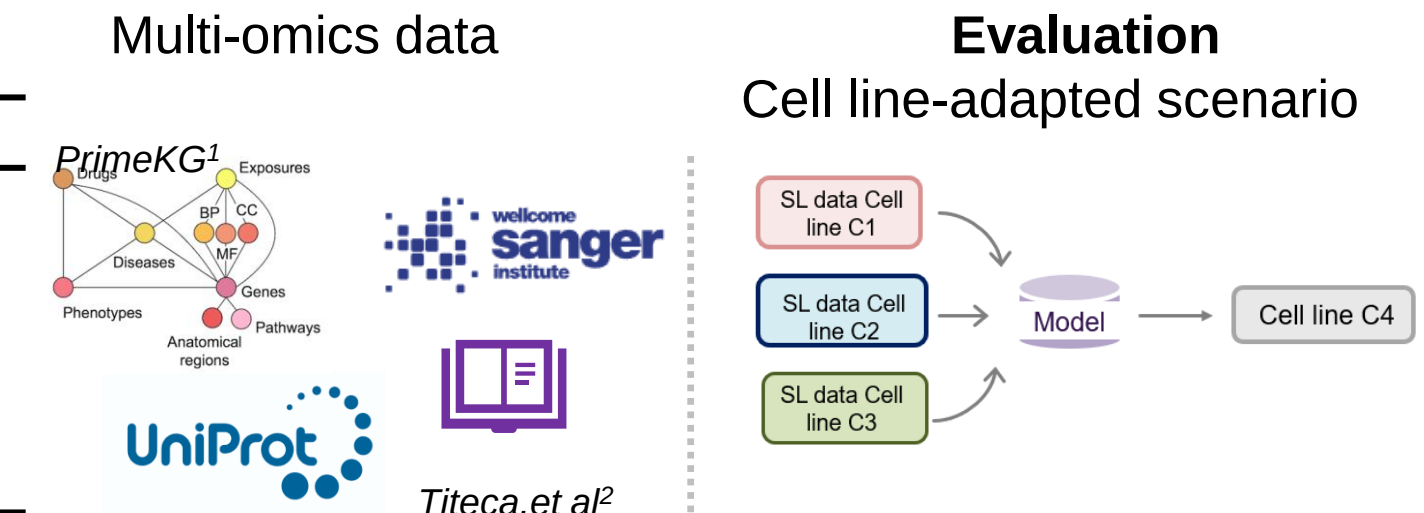
$$\mathcal{L}_{\text{Overall}} = \mathcal{L}_{\text{CE}} + \lambda_1 \cdot \mathcal{L}_{\text{CL}} + \lambda_2 \cdot \mathcal{L}_{\text{DL}}$$

Experiments and Results

Experimental Setup

Table 1. Statistics of six cell lines in our study.

Type of Cell Line	No. of positive	No. of negative	No. of gene	Labeled (%)
A375	22	110	142	1.32
A549	179	875	451	1.04
Jurkat	295	1,475	325	3.36
Mewo	284	1,024	1,345	0.14
22Rv1	209	737	44	100
Pk1	372	1,301	1,477	0.15



Results

Table 2. Performance comparison of our approach with the baselines (AUC score)

Model	Target cell line						Avg.
	A375	A549	Jurkat	Mewo	22Rv1	Pk1	
SLMGAE	0.6568±0.02	0.4270±0.00	0.6148±0.00	0.6075±0.00	0.3958±0.01	0.6418±0.00	0.5573
KG4SL	0.4359±0.02	0.4898±0.03	0.5181±0.05	0.5413±0.02	0.4642±0.05	0.5611±0.01	0.5016
PTGNN	0.4894±0.02	0.4332±0.02	0.4845±0.02	0.6410±0.04	0.3985±0.01	0.6035±0.01	0.5084
NSF4SL	0.6608±0.10	0.5008±0.05	0.5434±0.03	0.5865±0.02	0.4490±0.03	0.5324±0.02	0.5455
SLGNNCT	0.5551±0.04	0.5775±0.04	0.4830±0.05	0.6441±0.04	0.4557±0.06	0.6393±0.03	0.5591
MiT4SL	0.6781±0.05	0.6366±0.03	0.6200±0.00	0.6757±0.00	0.6137±0.02	0.6532 ± 0.00	0.6492

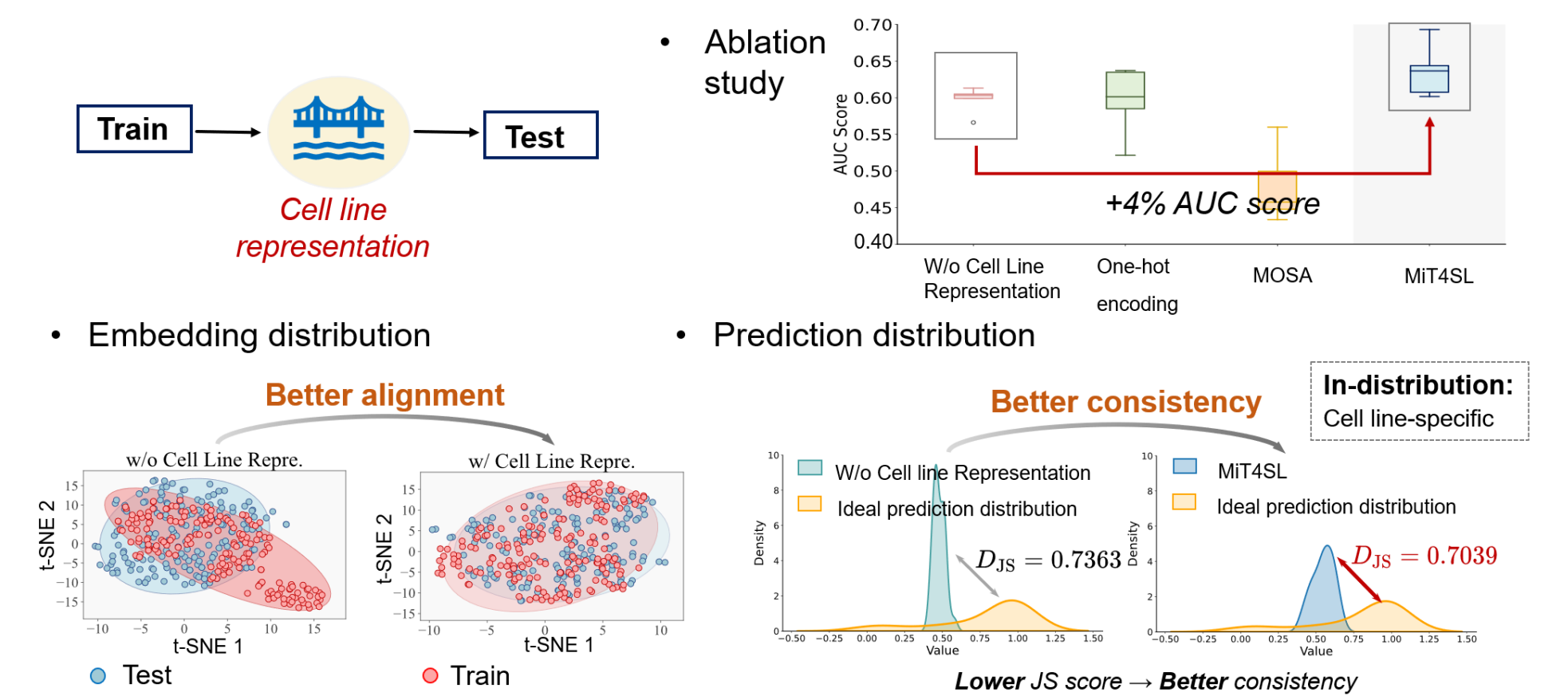
- MiT4SL achieves the best performance across various cell lines.
- MiT4SL achieves substantial improvements in challenging cell lines, e.g. 22Rv1 cell line.

Discussion

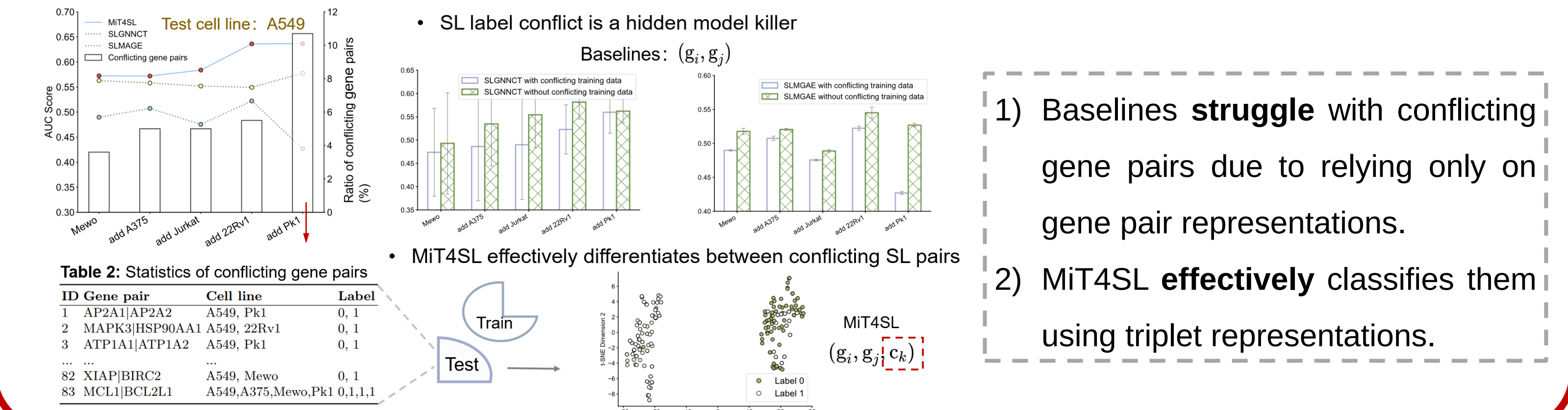
- Introducing cell line representations alleviates distribution shift

With Cell Line Representations:

- 1) +4% performance in the cell line-adapted scenario
- 2) Aligned embeddings (train vs. test)
- 3) Improved prediction consistency



- Introducing cell line representations address SL label conflicts



Conclusion

- We propose MiT4SL, a novel triplet representation model, designed to address data sparsity and label conflict across cell lines
- Comprehensive results show that MiT4SL significantly outperforms existing methods

■ **Applications:** Drug synergy prediction; Cell Line-Specific Target Discovery

References

- [1] Feng Y, Long Y, Wang H, et al. Benchmarking machine learning methods for synthetic lethality prediction in cancer[J]. Nature Communications, 2024, 15(1): 9058.
- [2] Chandak, P., Huang, K. & Zitnik, M. Building a knowledge graph to enable precision medicine. Sci Data 10, 67 (2023). <https://doi.org/10.1038/s41597-023-01960-3>.
- [3] Wang J, Wu M, Huang X, et al. SynLethDB 2.0: a web-based knowledge graph database on synthetic lethality for novel anticancer drug discovery[J]. Database, 2022, 2022: baac030.